

Geometry, Perimeter & Area — Level 2 — Answer Key

1. **C) 276 sq m** — Outer area = $30 \times 22 = 660$ sq m. A path 3 m wide running just inside the boundary leaves an inner rectangle of $(30-6) \times (22-6) = 24 \times 16 = 384$ sq m. Path area = $660 - 384 = 276$ sq m.
 2. **C) 444 sq m** — A path built outside adds 3 m all around, so the outer rectangle becomes $(40+6) \times (28+6) = 46 \times 34 = 1,564$ sq m. Inner (the park itself) = $40 \times 28 = 1,120$ sq m. Path area = $1,564 - 1,120 = 444$ sq m.
 3. **C) 616 sq cm** — Outer area = $50 \times 35 = 1,750$ sq cm. The photo itself (inner rectangle, 4 cm in from each side) = $(50-8) \times (35-8) = 42 \times 27 = 1,134$ sq cm. Frame area = $1,750 - 1,134 = 616$ sq cm.
 4. **C) 47 sq m** — Total area = $7 \times 5 + 4 \times 3 = 35 + 12 = 47$ sq m (the two sections' areas simply add, regardless of exactly how they're joined).
 5. **C) 110 sq cm** — Original area = $14 \times 9 = 126$ sq cm. Removing the 4×4 corner square removes 16 sq cm. Remaining area = $126 - 16 = 110$ sq cm.
 6. **B) 3 times** — Perimeter = $2(l+b)$. Making both sides 3 times as long: new perimeter = $2(3l+3b) = 3 \times 2(l+b) = 3 \times$ the original perimeter.
 7. **D) 16 times** — Area = side^2 . Making the side 4 times as long: new area = $(4 \times \text{side})^2 = 16 \times \text{side}^2 = 16 \times$ the original area.
 8. **C) 70 cm** — Six 5 cm squares in a row form a rectangle 30 cm long (6×5) and 5 cm wide. Perimeter = $2(30+5) = 70$ cm.
 9. **C) 192 sq cm** — Each tile has area $6 \times 4 = 24$ sq cm. Eight tiles placed with no gaps or overlaps always cover a total area of $8 \times 24 = 192$ sq cm, however they're arranged.
 10. **C) 1,536 sq m** — Field area = $50 \times 32 = 1,600$ sq m. Square plot area = $8 \times 8 = 64$ sq m. Uncovered area = $1,600 - 64 = 1,536$ sq m.
 11. **C) 46 cm** — The original rectangle's perimeter is $2(14+9) = 46$ cm. Cutting a rectangular notch out of a corner removes two short stretches of the original boundary but adds two new cut edges of exactly the same total length back in — so the perimeter of the L-shape is unchanged: still 46 cm.
 12. **B) 2** — The combined rectangle has a line of symmetry along the join between the two squares, and another line of symmetry running the other way through its centre — 2 in total.
 13. **C) 75°** — The angles of a quadrilateral always add up to 360° . Fourth angle = $360 - 80 - 95 - 110 = 75^\circ$.
 14. **C) 42** — The full grid has $8 \times 6 = 48$ unit squares. Removing a 3×2 block removes 6 unit squares. Remaining area = $48 - 6 = 42$ square units.
 15. **C) 48 cm** — Area 64 sq cm means each tile has side 8 cm (since $8 \times 8 = 64$). Two tiles side by side form a rectangle 16 cm by 8 cm. Perimeter = $2(16+8) = 48$ cm.
-

How to use this: 10+/15 is strong for a Level-2-focused set. Q11 (the corner-notch perimeter) is the trickiest idea here — it's worth re-deriving with pencil and paper: sketch the rectangle, cut the corner, and measure each remaining side yourself rather than just trusting the shortcut.